QP Code:- 25901384

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

Discipline Specific Course (DSC) – MG4DSCMAT200 – MATRIX ALGEBRA AND NUMBER THEORY

(2024 ADMISSION ONWARDS)

SCHEME FOR VALUATION

Duration: 2 Hours **Maximum Marks: 70**

Part A

Answer any **5** Questions.

Give maximum *2* marks for each question.

1. $3A = \begin{bmatrix} 3 & 0 & 3 \\ -3 & 0 & 3 \\ 0 & 3 & 3 \end{bmatrix}$ **…1 Mark**

   $-5A = \begin{bmatrix} -5 & 0 & -5 \\ 5 & 0 & -5 \\ 0 & -5 & -5 \end{bmatrix}$ **…1 Mark**

   $0A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ **…1 Mark**

2. If A is symmetric, then $A' = A$ **…1 Mark**

   If A is skew -symmetric, then $A' = -A$ **…1 Mark**

   $A = -A \Rightarrow A = 0$ **…1 Mark**

3. If $x_1, \dots, x_p$ are rows of matrix A then $x_1, \dots, x_p$ are linearly independent if the only scalars $\lambda_1, \dots, \lambda_p$ which are such that $\lambda_1 x_1 + \lambda_2 x_2 + \cdots + \lambda_p x_p = 0$ are $\lambda_1 = \lambda_2 = \cdots = \lambda_p = 0$.

   …2 Marks

   Example **…1 Mark**

4. For writing B is in normal form. **… 1 Mark**

   For getting normal form of A as B **…2 Marks**

   Rest **… 1 Mark**

   OR

   Getting rank A =1 **… 1 Mark**

   Rank B =1 **… 1 Mark**

   A and B are equivalent since they have the same rank **… 2 Marks**

5. For writing $100 = 25 \times 4 + 0$ **… 1 Mark**

   Since $25|100$, $\gcd(100, 25) = 25$ **… 2 Marks**

6. For writing any two prime numbers of the form $2^n + 1$ **…1 Mark**

   For writing a third prime number of the form $2^n + 1$ **…1 Mark**

[Eg: 3,5,17,257 etc]

7. For correct definition of complete system of residues modulo n **… 2 Marks**
   A collection of n integers $a_1, a_2, \dots, a_n$ is said to form a complete system of residues modulo n if every integer is congruent modulo n to one and only one of the a_k [in other words, if $a_1, a_2, \dots, a_n$ are congruent modulo n to $0,1,2, \dots, n-1$, taken in some order].
   For any correct example **… 1 Mark**
8. $\phi(10) = 4$ or $gcd(3,10) = 1$ **… 1 Mark**
   For getting $3^{\phi(10)} \equiv 1 \ (mod\ 10)$. Answer:1 **… 2 Marks**

Part B

Answer any **5** Questions.

Give maximum *6* marks for each question.

9. (a) Matrix form: $\begin{bmatrix} 0 & 1 & 4 \\ 1 & -2 & 1 \\ 0 & 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 23 \end{bmatrix}$ **…2 Marks**

   The augmented matrix is $\begin{bmatrix} 0 & 1 & 4 & 1 \\ 1 & -2 & 1 & 0 \\ 0 & 3 & -4 & 23 \end{bmatrix}$ **… 1 Mark**

   (b) For left multiplying an arbitrary matrix A with 3 rows by the matrix P **… 2 Marks**

   The effect of left multiplication by P is, in the product PA:

   Row 1 is the sum of Row 3 and α times Row 2 of A **… 1 Mark**

   Row 2 is β times Row 2 of A **… 1 Mark**

   Row 3 is the sum of Row 1 and γ times Row 2 of A **… 1 Mark**

10. Correct elimination below first pivot **… 2 Marks**

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 2 \\ 3 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 & 2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 1 & -2 & 1 & -2 \\ 0 & 1 & 2 & 1 & 2 \end{bmatrix} \begin{matrix} \\ \\ \rho_2 - \rho_1 \\ \rho_3 - 3\rho_1 \end{matrix}$$

Correct elimination below second pivot **… 2 marks**

$$\rightsquigarrow \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 & -3 \\ 0 & 0 & 3 & 1 & 1 \end{bmatrix} \begin{matrix} \\ \\ \rho_3 - \rho_2 \\ \rho_4 - \rho_2 \end{matrix}$$

Correct elimination below third pivot **… 2 marks**

$$\rightsquigarrow \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 & -3 \\ 0 & 0 & 0 & 4 & -8 \end{bmatrix} \begin{matrix} \\ \\ \\ \rho_4 + 3\rho_3 \end{matrix}$$

Rest **… 1 Mark**

Remark: Give full marks to any valid row-echelon form obtained through correct elementary row operations.

11. For writing $A_m = \begin{bmatrix} 1-m & -m \\ m & 1+m \end{bmatrix}$ **… 1 Mark**

For finding $A_n A_m$ **… 3 Marks**

For writing $A_{n+m} = \begin{bmatrix} 1-(n+m) & -(n+m) \\ n+m & 1+n+m \end{bmatrix}$ **… 2 Marks**

Rest **… 1 Mark**

12. For obtaining, row rank of $A = 2$ **… 3 Marks**

For obtaining, row rank of $B = 2$ **… 2 Marks**

A and B have the same Row rank **… 1 Mark**

13. Let $n, n+1, n+2$ and $n+3$ be the 4 consecutive integers **… 2 Marks**

Their product $P = n(n+1)(n+2)(n+3) = [n(n+3)][(n+1)(n+2]$ **… 1 Mark**

$= (n^2+3n)(n^2+3n+2)$ **… 1 Mark**

Put $k = n^2+3n$. k is an integer **… 1 Mark**

$P = k(k+2) = k^2+2k = (k+1)^2-1$ **… 2 Marks**

Rest **… 1 Mark**

14. Let $d = \gcd(a,b)$ **… 1 Mark**

First assume that exist integers x and y for which $c = ax+by$.

$d = \gcd(a,b) \Rightarrow d|a$ and $d|b \Rightarrow d|ax$ and $d|b\,y \Rightarrow d|ax+by \Rightarrow d|c \Rightarrow \gcd(a,b)\,|c$ **… 3 Marks**

Conversely assume that $\gcd(a,b)\,|c$.

$d|c \Rightarrow c = dk;\ k \in \mathbb{Z}$ **… 1 Mark**

$d = \gcd(a,b) \Rightarrow d = au+bv\,; u,v \in \mathbb{Z}$ **… 2 Marks**

$\Rightarrow c = a(ku)+b(kv); ku, kv \in \mathbb{Z}$ **… 2 Marks**

Let $x = ku$ and $y = kv$. Then $c = ax+by; x, y \in \mathbb{Z}$ **… 1 Mark**

15. For writing Fermat's theorem **… 1 Mark**

Since $7 \nmid a$, by Fermat's theorem $a^6 \equiv 1 (\text{mod } 7)$ **… 2 Marks**

$\Rightarrow 7|a^6-1$ **… 1 Mark**

$a^6-1 = (a^3)^2-1^2 = (a^3+1)(a^3-1)$ **… 2 Marks**

$\Rightarrow 7|(a^3+1)(a^3-1) \qquad \Rightarrow 7|(a^3+1)$ or $7|(a^3-1)$ **… 2 Marks**

16. For writing Wilson's theorem **… 1 Mark**

Since 29 is a prime, by Wilson's theorem $28! \equiv -1 (\text{mod } 29)$ **… 2 Marks**

$28! = 28 \times 27 \times 26!$ **… 1 Mark**

$28 \equiv -1 (\text{mod } 29)$ and $27 \equiv -2 (\text{mod } 29)$ **… 1 Mark**

$28! \equiv 2 \times 26!\ (\text{mod } 29)$ **… 1 Mark**

$2 \times 26! \equiv -1 (\text{mod } 29)$ **… 1 Mark**

$2 \times 26! \equiv 28 (\text{mod } 29)$ **… 1 Mark**

The remainder is 28 **… 1 Mark**

Part C

Answer any **3** Questions.

Give maximum *10* marks for each question.

17. (a) For proving the existence of additive identity **… 5 Marks**

(b) For proving the existence of additive inverse **… 5 Marks**

18. Coefficient matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 3 & \lambda \\ -3 & -4 & 1 \end{bmatrix}$ **… 2 Marks**

$$\rightsquigarrow \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & \lambda - 6 \\ 0 & -1 & 10 \end{bmatrix} \begin{matrix} \\ \rho_2 - 2\rho_1 \\ \rho_3 + 3\rho_1 \end{matrix}$$ **… 2 Marks**

$$\rightsquigarrow \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & \lambda - 6 \\ 0 & 0 & \lambda + 4 \end{bmatrix} \rho_3 + \rho_2$$ **… 2 Marks**

A non-trivial solution exists if and only if rank $A < 3$ **… 1 Mark**

$\lambda + 4 = 0 \Rightarrow \lambda = -4$ **… 1 Mark**

Then $A \rightsquigarrow \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & -10 \\ 0 & 0 & 0 \end{bmatrix}$ **… 1 Mark**

$y - 10z = 0 \Rightarrow y = 10z$ **… 1 Mark**

$x + y + 3z = 0 \Rightarrow x = -13\,z$

The solution is $x = -13\,z, y = 10z$, where z is arbitrary **… 2 Marks**

19. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & \alpha \\ 0 & 1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & \alpha \\ 0 & 1 & 1 \end{bmatrix} \rho_2 - \rho_1 \rightsquigarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 - \alpha \end{bmatrix} \rho_3 - \rho_2$ **… 2 Marks**

A is of rank 3 if and only if $\alpha \neq 1$. So A is invertible if and only if $\alpha \neq 1$ **… 2 Marks**

For getting $A^{-1} = \frac{1}{1-\alpha} \begin{bmatrix} 2 - \alpha & -1 & \alpha \\ -1 & 1 & -\alpha \\ 1 & -1 & 1 \end{bmatrix}$ **… 8 marks**

20. (a) For correct statement **… 3 Marks**

Division Algorithm:

Given integers a and b, with $b > 0$, there exist unique integers q and r satisfying $a = qb + r;\ 0 \leq r < b$.

(b) $q = -4$ and $r = 3$

$-25 = 7 \times (-4) + 3$ **… 3 Marks**

(c) By division algorithm a is of the form $3q, 3q + 1$ or $3q + 2$. **… 2 Marks**

If $a = 3q$, then $\frac{a(a^2+2)}{3} = q(9q^2+2)$, an integer **… 2 Marks**

If $a = 3q+1$, then $\frac{a(a^2+2)}{3} = \frac{(3q+1)((3q+1)^2+2)}{3} = (3q+1)(3q^2+2q+1)$, an integer **… 3 Marks**

If $a = 3q+2$, then $\frac{a(a^2+2)}{3} = \frac{(3q+2)((3q+2)^2+2)}{3} = (3q+2)(3q^2+4q+2)$, an integer **… 3 Marks**

21. (a)Any reasonable attempt **… 5 Marks**

(b) Definition of pseudoprime **… 1 Mark**

341=11×31 **… 1 Mark**

$2^{10} = 1024 = 31 \times 33 + 1 \Rightarrow 2^{10} \equiv 1(\text{mod } 31) \Rightarrow 2^{11} \equiv 2(\text{mod } 31)$ **… 3 Marks**

$2^{10} \equiv 1(\text{mod } 11) \Rightarrow 2^{30} = (2^{10})^3 \equiv 1(\text{mod } 11) \Rightarrow 2^{31} \equiv 2(\text{mod } 11)$ **… 3 Marks**

$2^{11\times 31} \equiv 2(\text{mod } 11 \times 31) \Rightarrow 2^{341} \equiv 2(\text{mod } 341)$ ⇒341 is a pseudoprime **… 3 Marks**

22. (a)For correct definition of Euler's phi function **… 2 Marks**

Euler's phi function: For $n \geq 1, \phi(n)$ denote the number of positive integers not exceeding n that are relatively prime to n. The function ϕ is called the Euler's phi function.

(b) If n is power of 2, say $n = 2^k$, with $k \geq 2$, then

$\phi(n) = \phi(2^k) = 2^k\left(1-\frac{1}{2}\right) = 2^{k-1}$, an even integer **… 3 Marks**

If n is not a power of 2, then we can write $n = p^k m$, where p is an odd prime, $k \geq 1$ and $gcd\left(p^k, m\right) = 1$ **… 2 Marks**

Then, $\phi(n) = \phi(p^k m) = \phi(p^k)\,\phi(m) = p^{k-1}(p-1)\phi(m)$, an even integer, since $p-1$ is even **… 3 Marks**

(c) $840 = 2^3 \times 3 \times 5 \times 7$ **… 2 Marks**

$\phi(840) = 840\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right) = \left(1-\frac{1}{7}\right) = 192$ **… 4 Marks**

N.B: Suitable stage marks may be fixed by the examiner for alternate correct answers.